Assignment-2: Forward & Inverse Kinematics and Workspace Analysis of DOBOT Magician Lite

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1 Abstract

This assignment focuses on performing forward and inverse kinematics for the Dobot Magician Lite and validating the results through both simulation and real-robot testing. A URDF model was created from CAD data to enable accurate simulation, inverse kinematics solving, and workspace analysis. The results confirmed that the developed kinematic model closely matches the behavior of the actual robot.

2 Introduction

Forward and inverse kinematics are essential for understanding and controlling robotic manipulators. In this assignment, these concepts are applied to the Dobot Magician Lite using geometric modeling and transformation matrices. A URDF-based digital model was built from CAD files to support simulation and analysis. The computed kinematics were validated in both simulation and on the physical robot to ensure accuracy and consistency.

3 Problem Statement

This assignment aims to:

- Derive and analyze forward and inverse kinematics for the Dobot Magician Lite.
- Create a URDF model from CAD data for simulation and workspace analysis.
- Validate the kinematic results on both a simulated model and the actual robot.

4 Forward Kinematics

Forward kinematics (FK) describes how joint angles determine the location and orientation of the robot's end-effector. For the Dobot Magician Lite, the FK model is constructed using homogeneous transformation matrices derived from the robot's geometry.

Each joint transformation is represented by the standard homogeneous transformation:

$$^{i-1}T_i = \begin{bmatrix} R_i & p_i \\ 0 & 1 \end{bmatrix}$$

where R_i is a 3×3 rotation matrix and p_i is a translation vector.

The complete forward kinematic equation from base to end-effector is:

$$^0T_4 = {}^0T_1 \cdot {}^1T_2 \cdot {}^2T_3 \cdot {}^3T_4$$

Rotation about the Z-axis for revolute joints is defined as:

$$R_z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Translation along an axis is represented as:

$$T_x(a) = \begin{bmatrix} 1 & 0 & 0 & a \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad T_z(d) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Dobot Simulation Interface

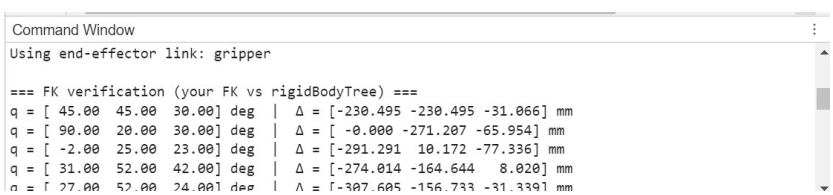

Figure 1: Dobot Magician Lite joint movement inside the simulation interface

By multiplying all link transformations, we obtain the final pose:

$$^0T_4 = \begin{bmatrix} n_x & o_x & a_x & p_x \\ n_y & o_y & a_y & p_y \\ n_z & o_z & a_z & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

which provides both end-effector orientation and position.

5 Inverse Kinematics

Inverse kinematics (IK) computes the joint angles $(\theta_1, \theta_2, \theta_3, \theta_4)$ for a desired end-effector position $P = (x, y, z)$.

Base Joint Angle

$$\theta_1 = \tan^{-1}\left(\frac{y}{x}\right)$$

Wrist Center

$$P_w = P - d_4 \cdot \hat{z}_{ee}$$
$$r = \sqrt{x^2 + y^2}, \quad h = z - d_1$$

Shoulder and Elbow Angles

$$D = \sqrt{r^2 + h^2}$$

$$\theta_3 = \pi - \cos^{-1}\left(\frac{a_2^2 + a_3^2 - D^2}{2a_2a_3}\right)$$

$$\theta_2 = \tan^{-1}\left(\frac{h}{r}\right) - \cos^{-1}\left(\frac{D^2 + a_2^2 - a_3^2}{2a_2D}\right)$$

Wrist Joint

$$\theta_4 = -(\theta_2 + \theta_3)$$

MATLAB FK Verification Output

6 Workspace Analysis

Workspace analysis evaluates the robot's reachable area and the quality of its movements using manipulability indices derived from the Jacobian matrix J:

$$\dot{x} = J(q)\dot{q}$$

Yoshikawa Manipulability Index

$$w = \sqrt{\det(JJ^T)}$$

Inverse-Condition Number

$$\kappa^{-1}(J) = \frac{\sigma_{\min}}{\sigma_{\max}}$$

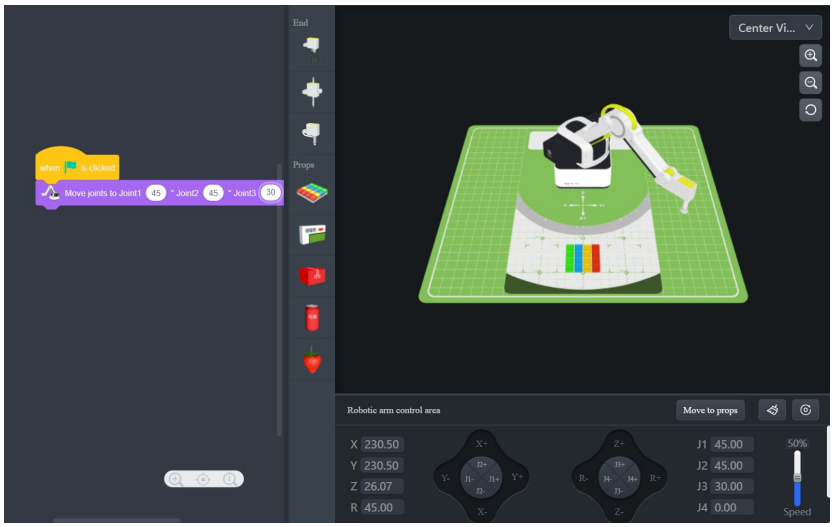

Figure 2: MATLAB script verifying forward kinematics using rigidBodyTree comparison

Asada Dynamic Manipulability Index

$$M = (JM_q^{-1}J^T)^{-1}$$

7 Conclusion

This assignment successfully implemented forward and inverse kinematics for the Dobot Magician Lite and validated the results through simulation and real hardware. The URDF model provided accurate simulation capability, and workspace analysis offered insights into the robot's dexterity and performance.